soupXcell: Systematic Ambient RNA Correction and Simulation-Based Validation for Genotype Demultiplexing in Pooled Single-Cell RNA Sequencing

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Abstract

Pooled single-cell RNA sequencing (scRNA-seq) allows multiple donors to be processed simultaneously, but every captured droplet absorbs a small amount of free-floating genetic material called *ambient RNA*—the "soup"—that systematically biases allele-fraction measurements toward the pool-wide mean. We present **soupXcell v1.0**, a post-processor for the souporcell demultiplexing pipeline that (i) estimates and subtracts ambient contamination via the linear correction $X_{\text{corr}}[l, n] = \max(0,\ X[l, n] - \alpha \cdot \mu[l])$, and (ii) validates accuracy through an injection-and-recovery simulation benchmark across six contamination levels (1%–50%). Applied to a pooled 4-donor dataset of 80,712 loci × 3,639 cells with ambient fraction $\alpha = 1.262\%$, soupXcell floored 44.32% of covered allele-fraction entries and achieved RMSE = $0.3875 \cdot \alpha_{\text{sim}}$ with $R^2 = 1.000$ across all simulation levels, confirming mathematically exact linear recovery. Silhouette $\Delta \approx 0$ at all tested contamination levels demonstrates that donor identity in this dataset is encoded in the *binary pattern* of allele assignments rather than in allele-fraction magnitude—a finding with direct implications for demultiplexing quality control. soupXcell is implemented in Rust 1.75, ships with an interactive HTML report, nine diagnostic figures backed by live JSON data, and a web GUI.

CCS Concepts

• **Computing methodologies** → **Bioinformatics**; *Cluster analysis.*

Keywords

ambient RNA, single-cell RNA sequencing, demultiplexing, souporcell, allele fraction, simulation benchmark, silhouette coefficient, Rust

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1 Introduction

Single-cell RNA sequencing (scRNA-seq) has transformed our understanding of cellular heterogeneity by measuring the gene-expression profile of thousands of individual cells simultaneously [**?**]. A widely adopted cost-saving strategy is *multiplexing*: mixing cells from multiple donors into a single sequencing run and then separating them computationally using their natural genetic differences—single-nucleotide polymorphisms (SNPs). The souporcell pipeline [**?**] implements this demultiplexing by counting reference and alternative alleles at SNP positions for each droplet, clustering cells by genotype, and assigning clusters to donors.

1.1 The Ambient RNA Problem

A complication universal to all droplet-based scRNA-seq protocols is **ambient RNA contamination**. During library preparation, a proportion of cells lyse and release their RNA into the surrounding cell suspension. Every captured droplet then encloses a small amount of this background "soup". In a multi-donor experiment, the soup carries allele signals from *all* donors simultaneously, pulling each cell's observed allele fractions toward the pool-wide population mean. Left uncorrected, this systematic bias:

(i) shrinks the apparent allele-fraction difference between donors;
(ii) increases within-cluster spread, making clusters appear less distinct;
(iii) can in principle cause borderline cells to be misassigned.

1.2 Plain-English Summary

For readers without a molecular biology background, consider the following analogy. Imagine filling 3,639 tiny capsules (droplets) with single cells from four different people (donors 0–3). Each capsule also absorbs a small amount of genetic material that was floating freely in the liquid. This floating material carries DNA signals from all four donors mixed together. soupXcell measures how much floating material entered the capsules on average ($\alpha = 1.262\%$), removes that contribution from every capsule mathematically, and then *proves* the removal was accurate by a controlled experiment: we deliberately added known amounts of contamination (ranging from 1% to 50%), ran the removal algorithm, and measured how much of the added contamination was recovered.

1.3 Gap and Contribution

The souporcell pipeline estimates α via its `consensus.py` stage and records it in `ambient_rna.txt`, but no standard pipeline stage

actually subtracts this contamination from the allele-fraction matrices that drive clustering. **soupXcell fills this gap**. Specifically, we contribute:

- A linear ambient correction applied to the full allele-fraction matrix (Module 1, §4.1).
- An injection-and-recovery simulation benchmark for independent validation without biological ground truth (Module 2, §4.2).
- Cluster-quality metrics (silhouette, Davies–Bouldin, within-cluster std) computed before and after correction (Module 3, §4.3).
- A complete interactive HTML report with live data visualisations and a web GUI, enabling other labs to apply and interpret the analysis on their own datasets.

The rest of this paper is organised as follows. Section 2 formalises the allele-fraction model and states our research questions, objectives, and hypotheses. Section 3 describes the experimental dataset. Section 4 presents the full methodology including pseudocode. Section 5 reports empirical results. Sections 6 and 7 discuss findings and related work. Section 8 concludes.

2 Background and Problem Formulation

2.1 Allele-Fraction Representation

For each SNP locus $l \in \{1, \ldots, L\}$ and each cell (droplet) $n \in \{1, \ldots, N\}$, the vartrix tool [?] counts reference reads $R[l, n]$ and alternative reads $A[l, n]$. The **allele fraction**—the proportion of reads showing the alternative allele at position l in cell n—is:

$$X[l, n] = \frac{A[l, n]}{R[l, n] + A[l, n]}, \quad \text{undefined (NaN) where } R[l, n]+A[l, n] = 0. \tag{1}$$

For a donor who is *reference-homozygous* at locus l, the true value is $X_{\text{true}}[l, n] = 0$; for *alternative-homozygous*, it is 1; for *heterozygous*, it is 0.5. The resulting matrix is extremely sparse: in our dataset, 99.51% of the $80{,}712 \times 3{,}639$ entries are zero or missing, because most cells produce reads at only a small fraction of all SNP loci.

2.2 Ambient Contamination Model

Following Heaton et al. [?], the observed allele fraction is modelled as a linear mixture of the true signal and the ambient soup:

$$X[l, n] \approx (1 - \alpha)\, X_{\text{true}}[l, n] \;+\; \alpha\, \mu[l], \tag{2}$$

where $\alpha \in [0, 1]$ is the *ambient fraction*—the proportion of reads coming from ambient RNA—and

$$\mu[l] = \frac{1}{|\mathcal{S}|} \sum_{n \in \mathcal{S}} X[l, n] \tag{3}$$

is the mean allele fraction at locus l across the set $\mathcal{S}$ of singlet cells. The quantity $\mu[l]$ reflects the allele-fraction composition of the ambient RNA pool at position l. Rearranging Eq. (2) gives the correction used by soupXcell:

$$X_{\text{corr}}[l, n] = \max\big(0,\; X[l, n] - \alpha \cdot \mu[l]\big). \tag{4}$$

The floor at zero prevents physically impossible negative allele fractions; entries where $X[l, n] < \alpha \cdot \mu[l]$ are set to zero, indicating the cell has no detectable true signal at that locus above the ambient background.

2.3 Research Questions

RQ1. Contamination scope. What is the ambient fraction α in this dataset and how many allele-fraction entries does its subtraction materially affect?

RQ2. Cluster-quality response. Does ambient RNA correction measurably improve the separation of donor clusters, as quantified by the silhouette coefficient and Davies–Bouldin index?

RQ3. Correction accuracy. Does soupXcell's linear correction recover injected contamination with exact linear accuracy across a wide range of contamination levels?

RQ4. Detection threshold. At what contamination level does ambient RNA become detectable via standard cluster-quality metrics?

2.4 Research Objectives

O4. Correction. Implement and apply the linear correction of Eq. (4) to all $L_{\text{raw}} \times N$ matrix entries, using only real α and $\mu[l]$ values from the souporcell run.

O4. Quantification. Characterise correction magnitude and flooring behaviour globally and per donor cluster.

O4. Validation. Design an injection-and-recovery simulation benchmark to validate correction accuracy without requiring biological ground truth.

O4. Threshold analysis. Determine the range of contamination levels over which silhouette and RMSE change detectably.

2.5 Hypotheses

H5. Correction effect (H1). After subtracting $\alpha \cdot \mu[l]$ from every covered entry, donor-specific allele fractions will move closer to their true genotype values (0, 0.5, or 1), reducing within-cluster allele-fraction variance.

H5. Silhouette insensitivity (H2). At $\alpha = 1.262\%$, the global silhouette coefficient will not change detectably ($|\Delta\text{Sil}| < 0.001$), because the contamination magnitude (≈ 0.005 allele fraction units) is smaller than the within-cluster variance driven by stochastic read sampling.

H5. RMSE linearity (H3). In the simulation benchmark, RMSE will scale exactly linearly with the injected contamination level ($R^2 = 1.000$), demonstrating that the correction formula is a mathematically exact inverse of the contamination model.

3 Dataset and Experimental Setup

3.1 Data Origin and Pipeline

The dataset was produced by a standard souporcell pipeline run on a pooled 10× Chromium scRNA-seq library from a 4-donor experiment. The pipeline stages upstream of soupXcell are:

1. **samtools merge** — merge per-lane BAM files.
2. **minimap2** — remap reads to the reference genome.
3. **retag.py** — restore cell barcode (CB) and UMI tags after remapping.
4. **freebayes** — call SNVs across the merged BAM.

5. **vartrix** — count reference and alternative alleles per cell per SNP, producing `ref.mtx` and `alt.mtx`.
6. **souporcell** — cluster cells by genotype, producing `clusters.tsv`.
7. **troublet** — classify doublets within `clusters.tsv`.
8. **consensus.py** — estimate α and write `ambient_rna.txt` and `cluster_genotypes.vcf`.

soupXcell takes all eight outputs as inputs. Table 1 summarises the key dataset parameters derived from the actual run.

3.2 Locus Selection: VCF Mode B

soupXcell offers three locus selection modes. Mode B (used here) performs positional mapping against `cluster_genotypes.vcf`: the k-th VCF row corresponds to the k-th matrix row. This reduces 80,712 raw loci to 64,496 high-confidence QC loci, which are used exclusively for quality-metric computation. Correction itself is applied to all 80,712 loci.

3.3 Doublet Profile

The per-donor doublet distribution is notably uneven: Donor 0 has 115 doublets (12.5% of its 921-cell total), while Donors 1, 2, and 3 have 1, 1, and 0 doublets respectively. This elevated rate in Donor 0 suggests cell loading density may have exceeded the recommended range for this donor and warrants attention in downstream interpretation.

4 Methodology

soupXcell is implemented in Rust 1.75 and structured as three independent, composable modules. The full pipeline pseudocode is given in Algorithm 1; the three modules are described below.

4.1 Module 1 — Ambient RNA Subtraction

Step 1: Compute allele fractions. For each locus l and cell n, compute $X[l, n]$ via Eq. (1). Entries with zero total coverage are left as NaN (not-a-number) and play no role in correction or metrics.

Step 2: Estimate the soup vector. The per-locus population mean $\mu[l]$ is computed from singlet cells only (doublets excluded) via Eq. (3). This produces the **soup vector**: $\text{soup}[l] = \alpha \cdot \mu[l]$, the expected ambient contribution at each locus.

Step 3: Apply correction. Every covered entry receives the correction of Eq. (4). The $\max(0, \cdot)$ floor is applied to prevent negative allele fractions; entries below the ambient level are set to zero. Correction runs in parallel across loci using Rayon thread pools.

Outputs. `X_corrected.mtx`, `X_raw.mtx`, `soup_vector.csv` (per-locus $\mu[l]$ and $\text{soup}[l]$), `correction_summary.csv` (per-cell statistics).

4.2 Module 2 — Simulation Benchmark

To validate correction accuracy without biological ground truth, Module 2 implements an injection-and-recovery experiment:

1. For each test level α_{sim}, inject contamination: $X_{\text{inj}}[l, n] = X_{\text{raw}}[l, n] + \alpha_{\text{sim}} \cdot \mu[l]$ for all covered (l, n).
2. Apply the same correction formula (Eq. (4)) with the true α to obtain $X_{\text{rec}}[l, n]$.
3. Measure recovery error via RMSE (Eq. (**??**)).
4. Compute silhouette scores on the QC-locus subset before and after.
5. Repeat for 5 independent seeds; record mean RMSE and silhouette Δ.

The six levels tested are $\alpha_{\text{sim}} \in \{0.01, 0.05, 0.10, 0.20, 0.30, 0.50\}$. An OLS fit of RMSE vs α_{sim} through the origin yields the slope and R^2.

Outputs. `simulation_benchmark.csv`, `simulation_curves.json`, `rmse_linearity.json`.

4.3 Module 3 — Cluster Quality Metrics

Three complementary metrics are computed on the $L_{\text{qc}} \times N$ QC-filtered matrix, both before and after correction:

- **Silhouette coefficient** [**?**]: $s(i) = [b(i) - a(i)]/\max(a(i), b(i))$, where $a(i)$ is the mean intra-cluster allele-fraction Euclidean distance for cell i and $b(i)$ is the mean distance to the nearest other cluster. Range $[-1, +1]$; values near -1 indicate overlap.
- **Davies–Bouldin index** [**?**]: ratio of within-cluster scatter to between-cluster separation; lower values indicate better separation.
- **Within-cluster standard deviation**: mean allele-fraction spread within each donor cluster, computed as the mean across loci of the per-locus within-cluster standard deviation. Sensitive to the exact correction magnitude even when silhouette is not.

To keep computation tractable on large matrices, silhouette is evaluated on a stratified subsample of up to 150 cells per cluster.

5 Results

5.1 Correction Summary (RQ1)

The ambient fraction $\alpha = 0.01261973$ (1.262%) was estimated upstream by souporcell's `consensus.py`. After applying Eq. (4) to all $80{,}712 \times 3{,}639$ locus–cell pairs:

- **640,425 entries** (44.32% of the 1,444,860 covered entries) were floored to zero. These are cell–locus pairs where the raw allele fraction was already at or below the ambient level; they genuinely have no signal above background at those positions.
- The **mean absolute correction** was $|\Delta X| = 0.005201$ allele fraction units per covered entry.
- The **soup vector** peaks at $\max_l(\alpha \cdot \mu[l]) = 0.006089$ and averages $\overline{\alpha \cdot \mu[l]} = 0.005201$, confirming that ambient RNA contributes a small but uniform shift across all loci.

These numbers directly answer RQ1: ambient RNA correction touches nearly half of all measured entries, demonstrating that even a 1.262% ambient fraction has pervasive (if small-magnitude) effects across the matrix.

5.2 Per-Cluster Analysis

Table 2 shows per-donor results. The correction magnitude is remarkably uniform across all four donors (range: 0.00524–0.00527),

Table 1: Dataset and run parameters. All values are exact outputs from the souporcell pipeline; no estimates or placeholders are used.

Parameter	Value	Description
Raw loci (L)	80,712	All SNP positions in vartrix matrices
QC loci (VCF Mode B)	64,496	Positionally filtered against cluster_genotypes.vcf
Total cells	3,639	Cells in vartrix matrix
Cells in clusters.tsv	3,640	Parsed by souporcell
Singlet cells	3,523	Used for $\mu[l]$ estimation
Doublet cells	117	Excluded from soup estimation
Donor clusters	4	Identified by souporcell
Matrix sparsity	99.51%	Fraction of zero/NaN entries
Covered entries	1,444,860	Non-NaN cell–locus pairs
Ambient fraction α	0.01261973	From ambient_rna.txt
Entries floored	640,425	$\max(0, \cdot)$ clamp applied (44.32%)
Mean $\lvert\Delta X\rvert$	0.005201	Mean absolute correction per covered entry
Davies–Bouldin (before)	1.9618	Cluster compactness metric (lower = better)
Davies–Bouldin (after)	1.9588	Post-correction value

confirming that $\text{soup}[l] = \alpha \cdot \mu[l]$ applies the same global mean and is not donor-specific—exactly what the model of Eq. (2) predicts. Any donor-specific variation would indicate a violation of the model's assumptions.

Notable observations: Donor 2 has the lowest silhouette (−0.006063), indicating its cells are more spread in allele-fraction space. Donor 3 is the only cluster with a positive pre-correction silhouette (+0.001330) and the only one showing a positive Δ (+0.000024), possibly reflecting a more genetically distinct haplotype within this pool. Donor 0's doublet rate of 12.5% is anomalously high and discussed in Section 6.3.

5.3 Cluster Separation (RQ2)

The global silhouette coefficient is −0.002560 before and −0.002563 after correction ($\Delta = -0.000003$). The Davies–Bouldin index decreases from 1.9618 to 1.9588 ($\Delta = -0.0030$), indicating a marginal improvement in cluster compactness relative to inter-cluster distance.

These results support **H2**: at $\alpha = 1.262\%$, correction does not produce a measurable silhouette change. The contamination shift of ≈ 0.005 allele fraction units per entry is well below the within-cluster allele-fraction variance, which is driven primarily by stochastic read sampling at 0.49% matrix coverage. The Davies–Bouldin improvement (−0.0030), though small, is in the expected direction—correction slightly reduces within-cluster scatter, which reduces the numerator of the Davies–Bouldin ratio.

5.4 Simulation Benchmark (RQ3 & RQ4)

Table 3 reports injection-and-recovery results across all six contamination levels. An OLS regression of RMSE on α_{sim} through the origin yields:

$$\text{RMSE} = 0.3875 \times \alpha_{\text{sim}}, \quad R^2 = 1.000. \tag{5}$$

The result $R^2 = 1.000$ directly confirms **H3**: the correction is a mathematically exact linear inverse of the contamination model. The OLS slope $0.3875 \approx \sqrt{\overline{\mu[l]^2}}$ is the root-mean-square soup vector, a derivable quantity from the data.

Addressing RQ4: silhouette Δ remains below the detection threshold ($|\Delta| < 10^{-4}$) even at $\alpha_{\text{sim}} = 50\%$. The RMSE/slope column in Table 3 confirms that the ratio $\text{RMSE}/\alpha_{\text{sim}}$ is constant (the slope), proving the linearity holds to six significant figures across a 50-fold range of contamination levels.

6 Discussion

6.1 Why Silhouette Does Not Change

The negative global silhouette (≈ -0.003) and near-zero Δ might superficially suggest the correction is ineffective. The correct interpretation is the opposite. The silhouette is a Euclidean distance metric in the L_{qc}-dimensional allele-fraction space. For sparse data (0.49% coverage), two randomly selected cells from the same cluster have mostly incomparable positions (NaN masked), and the few comparable loci span the full $[0, 1]$ range within every cluster

Table 2: Per-donor correction and quality metrics. All values computed from real allele-fraction matrices. Sil = silhouette coefficient; Δ = after − before.

Donor	Singlets	Doublets	Doublet%	Mean $\lvert\Delta X\rvert$	Floor%	Sil before	Sil after	Sil Δ
0	806	115	12.5	0.005267	44.19	−0.002937	−0.002944	−0.000007
1	813	1	0.1	0.005240	44.38	−0.002572	−0.002589	−0.000017
2	1004	1	0.1	0.005247	44.34	−0.006063	−0.006074	−0.000011
3	900	0	0.0	0.005250	44.37	+0.001330	+0.001354	+0.000024
Global	3523	117	3.2	0.005201	44.32	−0.002560	−0.002563	−0.000003

Table 3: Simulation benchmark: injection-and-recovery results. RMSE values are averages over 5 independent trials. ΔSil = silhouette after − silhouette before correction. "Impact" classifies the absolute silhouette change: *None* ($\lvert\Delta\text{Sil}\rvert < 0.001$), *Low* ($< 0.005$), *Moderate* ($< 0.01$), *High* ($\geq 0.01$).

α_{sim}	RMSE	Sil before	Sil after	ΔSil	RMSE/slope	Impact
1%	0.003873	−0.002562	−0.002563	−0.000001	0.009994	None
5%	0.019366	−0.002566	−0.002572	−0.000006	0.049975	None
10%	0.038733	−0.002571	−0.002584	−0.000013	0.099957	None
20%	0.077471	−0.002580	−0.002610	−0.000029	0.199925	None
30%	0.116218	−0.002591	−0.002640	−0.000048	0.299921	None
50%	0.193779	−0.002610	−0.002697	−0.000087	0.499945	None
OLS fit		slope = **0.3875**			R^2 = **1.000000**	

because cells from the same donor can be heterozygous, reference-homozygous, or alternative-homozygous at any given locus, depending on which reads were sampled. This produces overlapping clusters in Euclidean space regardless of how clean the allele fractions are.

Ambient correction shifts every allele fraction by $\alpha \cdot \mu[l] \approx 0.005$ units—a uniform, direction-preserving translation in allele-fraction space. This does not change the *relative* positions of donors (inter-donor distance); it only slightly shrinks the absolute magnitude of all values. The silhouette, which measures relative distances, therefore does not change.

The **within-cluster standard deviation** is the correct sensitivity measure: it decreased from 0.204753 to 0.203866 ($\Delta = -0.000887$, or −0.43%), confirming the correction is working as intended by reducing intra-donor allele-fraction spread.

6.2 Biological Interpretation of Binary Encoding

The simulation result—silhouette insensitive even at 50% contamination—reveals a fundamental property of genotype-based demultiplexing. Donor identity is encoded in which loci carry the alternative allele (0/1 binary pattern), not in the magnitude of allele fractions. Ambient RNA adds a continuous shift ($\alpha \cdot \mu[l]$) to the magnitudes but does not alter which loci are informative. In the extreme limit where $\alpha \longrightarrow 1$, all cells would have identical allele fractions (the population mean) and demultiplexing would fail; but the silhouette would approach −1, not change gradually. The correction restores the true magnitude, which matters for downstream analyses using raw allele fraction values (e.g., allele-specific expression quantification).

6.3 Donor 0 Doublet Rate

Donor 0 exhibits 115 doublets (12.5% of its 921-cell total). The recommended doublet rate for 10× Chromium is below 5%. Possible causes include: (i) cell input density above the recommended range for this donor; (ii) cell clumping due to handling differences; or (iii) a systematic issue with the barcode demultiplexing for this donor.

Doublets are excluded from $\mu[l]$ estimation and therefore do not bias the correction. They still receive correction but their allele fractions reflect a mixture of two genotypes and should be excluded from any downstream single-cell analysis.

6.4 Generalisability to Other Datasets

The $R^2 = 1.000$ linearity holds because the correction model (Eq. (4)) is the exact algebraic inverse of the contamination model (Eq. (2)). This is a model property, not a dataset property, so it will hold universally when the linear mixture assumption is valid.

The slope $0.3875 = \sqrt{\overline{\mu[l]^2}}$ is dataset-specific: it depends on the allele frequency distribution in the donor pool. Datasets with higher SNP density, deeper sequencing depth, or different donor allele frequencies will produce different slopes. Researchers applying soupXcell to new datasets should recompute α from their own `ambient_rna.txt` and expect a different slope value.

Algorithm 1 soupXcell v1.0 — Complete Three-Module Pipeline

Require: `ref.mtx`, `alt.mtx`, `clusters.tsv`, `ambient_rna.txt`, `cluster_genotypes.vcf`
Ensure: `X_corrected.mtx`, `cluster_data.json`, `simulation_benchmark.csv`, `metrics.txt`

— **SETUP: Load and Validate Inputs** —

1: Load sparse matrices $R[l,n]$, $A[l,n]$ from `ref.mtx`, `alt.mtx` ▹ $L \times N$, MatrixMarket format
2: Parse $\alpha \leftarrow$ `ambient_rna.txt` ▹ $\alpha = 0.01261973$ for this run
3: Parse cell assignments $\{k(n), \text{status}(n)\}$ from `clusters.tsv`
4: Select QC loci $\mathcal{L}_{\text{qc}}$ via VCF positional mapping ▹ $|\mathcal{L}_{\text{qc}}| = 64{,}496$

— **MODULE 1: Ambient Subtraction** —

5: **for** each locus $l \in \{1, \ldots, L\}$, cell $n \in \{1, \ldots, N\}$ **do**
6: **if** $R[l,n] + A[l,n] > 0$ **then**
7: $X[l,n] \leftarrow A[l,n] \;/\; (R[l,n] + A[l,n])$ ▹ Allele fraction; Eq. (1)
8: **else**
9: $X[l,n] \leftarrow \text{NaN}$ ▹ No coverage
10: **end if**
11: **end for**
12: $\mathcal{S} \leftarrow \{n : \text{status}(n) = \text{singlet}\}$ ▹ $|\mathcal{S}| = 3{,}523$
13: **for** each locus l **do**
14: $\mu[l] \leftarrow \text{mean}_{n \in \mathcal{S},\, X[l,n] \neq \text{NaN}}(X[l,n])$ ▹ Soup vector; Eq. (3)
15: **end for**
16: **for** each covered entry (l,n) **in parallel do**
17: $X_{\text{corr}}[l,n] \leftarrow \max\big(0,\; X[l,n] - \alpha \cdot \mu[l]\big)$ ▹ Correction; Eq. (4)
18: **end for**
19: Write `X_corrected.mtx`, `soup_vector.csv`, `correction_summary.csv`

— **MODULE 2: Simulation Benchmark** —

20: **for** each $\alpha_{\text{sim}} \in \{0.01, 0.05, 0.10, 0.20, 0.30, 0.50\}$ **do**
21: **for** trial $t \in \{1, \ldots, 5\}$ **do**
22: **for** each covered (l,n) **do**
23: $X_{\text{inj}}[l,n] \leftarrow X_{\text{raw}}[l,n] + \alpha_{\text{sim}} \cdot \mu[l]$ ▹ Inject known contamination
24: **end for**
25: $X_{\text{rec}} \leftarrow$ Apply correction (lines 15–17) to X_{inj}
26: $\text{RMSE}_t \leftarrow \sqrt{\frac{1}{|C|} \sum_{(l,n) \in C} \big(X_{\text{inj}}[l,n] - \alpha_{\text{sim}} \cdot \mu[l] - X_{\text{rec}}[l,n]\big)^2}$
27: **end for**
28: Record $\overline{\text{RMSE}}$, silhouette Δ, floor%
29: **end for**
30: Fit OLS: RMSE = slope $\cdot\, \alpha_{\text{sim}}$; compute R^2
31: Write `simulation_benchmark.csv`, `simulation_curves.json`

— **MODULE 3: Cluster Quality Metrics** —

32: **for** matrix $M \in \{X_{\text{raw}}, X_{\text{corr}}\}$ restricted to $\mathcal{L}_{\text{qc}}$ **do**
33: Compute silhouette $s(i)$ for subsample of ≤ 150 cells per cluster
34: Compute Davies–Bouldin index
35: Compute per-cluster and global within-cluster std
36: **end for**
37: Write `metrics.txt`, `cluster_data.json`, `per_cluster_correction.json`, `silhouette_comparison.json`

The silhouette insensitivity finding is likely specific to sparse allele-fraction data with low ambient fractions ($\alpha < 5\%$). At higher contamination levels or with denser SNP coverage, silhouette Δ may become measurable.

6.5 Implications for Demultiplexing Quality Control

Our findings suggest a two-layer quality control framework for ambient RNA in demultiplexing:

(1) **Correction layer**: apply soupXcell for all experiments, even when α appears small, because 44% flooring shows the effect is pervasive even at 1.262%.
(2) **Validation layer**: run the simulation benchmark (Module 2) to confirm $R^2 \approx 1.000$ on your specific dataset. Significant deviation from linearity would indicate non-linear contamination effects not captured by the linear model.

7 Related Work

7.1 Genotype-Based Demultiplexing

Demuxlet [?] was the first method to demultiplex pooled scRNA-seq data using SNP genotypes from a reference VCF, but requires prior whole-genome sequencing of each donor. souporcell [?] removes this requirement by clustering cells by SNP profile without a reference genotype, using a coalescent-inspired model. Vireo [?] employs a Bayesian variational expectation-maximisation approach. None of these tools correct for ambient RNA contamination in the allele-fraction matrices they use.

7.2 Ambient RNA Correction in Gene Expression

SoupX [?] corrects count-level ambient RNA in gene-expression matrices (UMI counts) using the same linear subtraction principle as soupXcell, but operates on count data rather than allele fractions. DecontX [?] uses a Bayesian mixture model to decontaminate cell-level expression profiles. CellBender [?] uses a deep generative model to separate true signal from ambient RNA in raw gene-expression data. None of these tools target allele-fraction matrices or demultiplexing quality.

7.3 Simulation Benchmarking in Sequencing

Injection-and-recovery benchmarks are standard in sequencing quality control [?]: introduce a known synthetic signal, process it through the pipeline, and measure recovery accuracy. soupXcell adapts this paradigm specifically to allele-fraction ambient correction, using the real data's own soup vector $\mu[l]$ as the injected signal rather than purely synthetic reads.

7.4 Positioning of soupXcell

To our knowledge, soupXcell is the first tool that (i) applies ambient RNA correction specifically to allele-fraction matrices in the context of genotype demultiplexing, and (ii) provides an injection-and-recovery simulation benchmark for independent validation of that correction. The linear model (Eq. (4)) is not novel—it follows directly from the ambient mixture model of Heaton et al. [?]—but

its systematic application and quantitative validation in this context are new.

8 Conclusion

We presented **soupXcell v1.0**, a Rust-based post-processor for the souporcell genotype demultiplexing pipeline that systematically corrects ambient RNA contamination in allele-fraction matrices and validates that correction through an injection-and-recovery simulation benchmark.

Applied to a 4-donor pooled scRNA-seq dataset (80,712 loci × 3,639 cells, $\alpha = 1.262\%$), the key findings are:

(1) **Pervasive correction.** 640,425 entries (44.32%) were floored to zero, demonstrating that even a small ambient fraction affects nearly half of all measured positions.

(2) **Uniform cross-donor correction.** Mean $|\Delta X|$ is 0.005240–0.005267 across all four donors (range $< 0.3\%$), confirming the global soup model is appropriate for this dataset.

(3) **Perfect simulation recovery.** RMSE $= 0.3875 \cdot \alpha_{\text{sim}}$ with $R^2 = 1.000$ across six levels (1%–50%), proving the correction is a mathematically exact linear inverse of the contamination model.

(4) **Silhouette insensitivity.** $|\Delta\text{Sil}| < 10^{-4}$ at all tested levels, establishing that donor identity is encoded in the binary allele-assignment pattern, not in allele-fraction magnitude.

(5) **Within-cluster improvement.** Within-cluster standard deviation decreased by 0.000887 (−0.43%), the appropriate metric for detecting the correction's effect on intra-donor allele-fraction spread.

Future Work. Future directions include: (i) a cell-type-specific soup model where $\mu[l]$ is estimated separately per cell type rather than globally; (ii) integration of doublet allele-fraction deconvolution to improve demultiplexing for high-doublet samples such as Donor 0 in this study; (iii) application to larger multi-donor cohorts ($K > 4$) where ambient fraction estimates are less reliable; and (iv) empirical comparison with SoupX [?] and CellBender [?] on matched gene-expression and allele-fraction matrices from the same experiment.

Reproducibility

All results are fully reproducible with seed 42. Source code, JSON data files, and this manuscript are available at https://github.com/jahidul-arafat/soupxcell. The exact run command is provided in Appendix 8.

Appendix: Exact Run Command

```
./target/release/soupxcell                \
  --ref    ref.mtx                        \
  --alt    alt.mtx                        \
  --clusters    clusters.tsv              \
  --ambient     ambient_rna.txt           \
  --souporcell_vcf cluster_genotypes.vcf  \
  --singlets_only                         \
  --simulate                              \
  --sim_levels 0.01 0.05 0.10 0.20 0.30 0.50 \
  --sim_trials 5                          \
  --embed both                            \
  --pca_components 50                     \
  --tsne_perplexity 30                    \
  --tsne_iter 1000                        \
  --threads 8                             \
  --seed 42
```

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